

Who Shapes Cybersecurity Conversation on Bluesky?

Social Web Analytics (COMP3020): Group Project Report

Group 32

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Code and data repository: <https://github.com/Yousuf963/comp3020-group32>

1 Research Questions

This project examines who shapes cybersecurity conversation on Bluesky by analysing posts containing “cybersecurity” or “infosec” and a follow network surrounding a prominent cybersecurity account.

1. **Text (2.3):** What are the most frequently discussed topics in cybersecurity-related posts on Bluesky?
2. **Clustering (2.4):** What groups of similar topics can be identified in cybersecurity-related Bluesky posts?
3. **Hypothesis (2.2):** Are posts from automated (bot) accounts less likely to receive engagement than posts from human managed accounts?
4. **Network (2.5):** In the follow network around the most-followed active author in the cybersecurity discussion, which accounts are most central, what kind of accounts are they, and does this network belong to the cybersecurity community?

2 Analyses

2.1 Data Collection

Source and access. the data (post) were collected from Bluesky using the `atrrr` package, which accesses the public Bluesky (AT Protocol) API after authenticating with an app password. The Query searched for the latest posts related to two queries namely "cybersecurity" and "infosec", requesting up to 3,000 posts each with `search_post()`. Data were collected once on 06 October 2026 (AEDT) and frozen; The results of this report comes from that single snapshot from Bluesky. The collection script is `scripts/Collect.R` in the repository; the report never calls the API.

R packages. that used in this report: `atrrr` (data collection), `tidyverse` (cleaning and wrangling), `knitr` (tables). Later sections use `tm`, `wordcloud` and `igraph`.

Volume and time window. 6188 posts were collected, posted between 30 Sep and 05 Oct 2026 (UTC).

Table 1: Variables in the cleaned post dataset

Variable	Description
id (uri)	Unique post identifier
created_at	Time the post was indexed by Bluesky (UTC)
author	Author handle
text	Post text
query	Search query that returned the post
like_count, repost_count, reply_count, quote_count	Engagement counts at collection time
has_link, is_reply, n_words, n_mentions	Derived post characteristics

Cleaning. The Table below shows how many posts each step removed. There are some Duplicates matched in both queries. Only the posts tagged as English were taken, so the text analysis uses one language. To detect bot accounts, we flagged every handle containing “bot” (13 accounts) and manually reviewed and verified each Bluesky profile: 9 were confirmed bot (automated) accounts (vulnerability alerts, news/RSS feeds and mirror accounts) and removed; 4 were real people and organisations were kept. Finally, posts with fewer than 5 words were removed as too short to analyse.

Table 2: Posts remaining and removed at each cleaning step

step	posts	removed
Collected	6188	–
After removing duplicates	4968	1220
After English-only filter	2845	2123
After removing bot accounts	1331	1514
After removing posts < 5 words	1293	38

The final dataset contains **1293 posts from 574 authors**. Notably, bot accounts wrote 1514 of the 2845 English posts (53%), almost all of them from a single vulnerability-alert feed (euvd-bot), which motivates the hypothesis test in Section 2.2.

Follow network. A seed user has selected in the process from the topic’s authors : among authors who posted at least 3 times (excluding bots) and follow at least 30 accounts, the seed is the one with the most followers (sifu.tweety.fish). These filters ensures the seed is the most-followed author so that it justifies an active voice in the cybersecurity conversation rather than an automated feed, and follows enough accounts to build a usable network; the most-followed author overall followed only a handful of accounts. The query retrieved 59 of the 985 accounts the seed follows, and for each of these, the accounts they follow (between 1 and 99 per friend, as returned by the API), giving 3619 directed “follows” edges. 0 retrievals failed.

Limitations.

- *Sample:* The search query only returns the most recent posts, therefore, the data only covers five days and reflect that week’s news.

- *Keyword search*: posts related to cybersecurity that do not use keywords are missed, and some matching posts are irrelevant and off topic.
- *Language filter*: posts that are not in English language were removed, Only English posts are included.
- *Bot detection*: bots were identified by handle and manual review only; automated accounts without “bot” in the handle may remain.
- *Engagement counts* were recorded at collection time, so the newest posts had less time to gather likes.
- *Network*: the follow network is a two-step sample around one seed with capped limits, not the full Bluesky network. Mentions were only 3% of the posts very less to build a mention network.

2.2 Hypothesis Testing

Research question. *Are posts from automated (bot) accounts less likely to receive engagement than posts from human managed accounts?*

The data findings at Section 2.1 showed that automated account(bots) produce over half of the English cybersecurity posts. This test examines whether that volume helps grabbing more attention from the users or not.

The process used English posts **before** bot removal, applying the same duplicate, language and length filters as the cleaned data, and label each post by account type.

```
# Same bot list as scripts/Clean.R (13 flagged handles, 4 manually kept as real accounts)
flagged <- unique(posts_raw$author_handle[str_detect(posts_raw$author_handle, "bot")])
kept    <- c("ebottcher.bsky.social", "repeatablerobot.bsky.social",
            "adybot.bsky.social", "progressiverobot.bsky.social")
bots    <- setdiff(flagged, kept)

hyp <- posts_raw %>%
  distinct(uri, .keep_all = TRUE) %>%
  filter(map_lgl(langs, ~ "en" %in% unlist(.x))) %>%
  mutate(n_words      = str_count(str_squish(text), "\\S+"),
         account_type = if_else(author_handle %in% bots, "Bot", "Human"),
         engaged       = if_else(like_count + repost_count + reply_count > 0,
                                "Engaged", "No engagement")) %>%
  filter(n_words >= 5)

tab <- table(hyp$account_type, hyp$engaged)
tab
```

```
##
##           Engaged No engagement
## Bot           219           1295
## Human          730           563
```

```
round(prop.table(tab, margin = 1), 3)      # proportion engaged within each group
```

```
##
##           Engaged No engagement
##   Bot      0.145      0.855
##   Human    0.565      0.435
```

```
hyp %>% filter(account_type == "Bot") %>%
  group_by(author_handle) %>%
  summarise(posts = n(), share_engaged = round(mean(engaged == "Engaged"), 3)) %>%
  arrange(desc(posts)) %>%
  kable(caption = "Posts and engagement for each bot account")
```

Table 3: Posts and engagement for each bot account

author_handle	posts	share_engaged
euvd-bot.bsky.social	1499	0.137
judgement-bot.bsky.social	11	1.000
newsbot.app.leafplaza.eu	3	0.667
google.xmirror.bot	1	1.000

```
hyp %>%
  count(account_type, engaged) %>%
  group_by(account_type) %>% mutate(prop = n / sum(n)) %>%
  filter(engaged == "Engaged") %>%
  ggplot(aes(account_type, prop)) +
  geom_col() +
  scale_y_continuous(labels = scales::percent) +
  labs(x = NULL, y = "Posts with >= 1 like, repost or reply")
```

Hypotheses.

- H0: account type and receiving engagement are independent.
- H1: account type and receiving engagement are associated.

Why this test. Since both variables are categorical (bot/human; engaged/not), therefore, chi-squared test of independence is appropriate. Engagement counts are uneven (the median post receives 0 likes), so comparing means with a t-test would be misleading; converting to “any engagement” avoids this. The test requires expected counts of at least 5 in every cell, checked below. Following the randomisation approach, the p-value is simulated: account type and engagement are shuffled 2,000 times to break any dependency, and the observed statistic is compared with the shuffled values.

```
set.seed(3020)      # makes the simulation reproducible
test <- chisq.test(tab, simulate.p.value = TRUE, B = 2000)
test$expected      # all should be >= 5
```

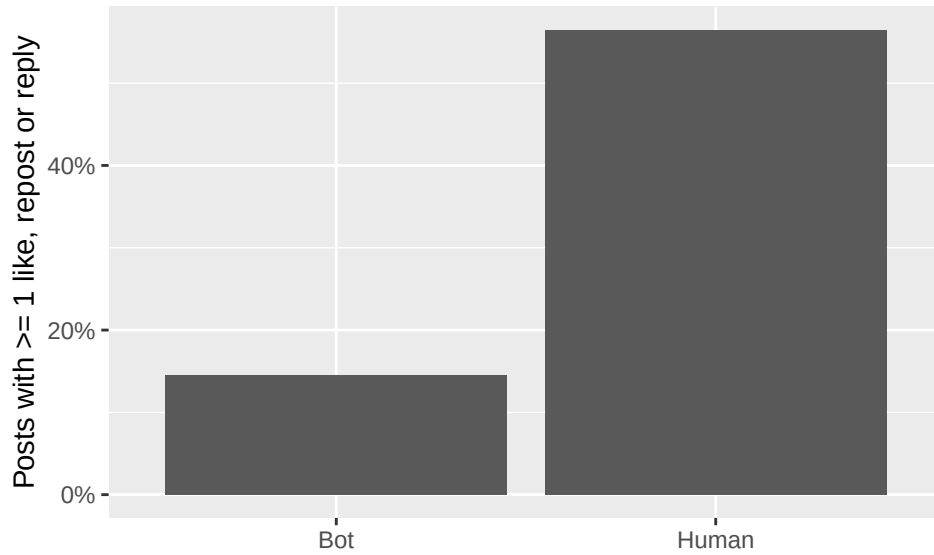


Figure 1: Share of posts receiving any engagement, by account type

```
##
##           Engaged No engagement
## Bot      511.8582    1002.1418
## Human    437.1418     855.8582
```

```
test
```

```
##
## Pearson's Chi-squared test with simulated p-value (based on 2000
## replicates)
##
## data:  tab
## X-squared = 549.55, df = NA, p-value = 0.0004998
```

Result and interpretation. From the observation of the statistic it can be said that The ($\chi^2 = 549.55$) was larger than all the shuffled 2,000 values from the tables, giving a simulated p-value of about 0.0005, which is below $\alpha = 0.05$, so we reject H_0 : means the account type and receiving engagement are associated. 56.5% of posts from the human managed accounts received at least one like, repost or reply, compared with 14.5% of posts from bot accounts. However, the per-account table shows that 1,499 of the 1,514 automated (bot) posts come from a single vulnerability-alert feed, euvd-bot, while the other three automated accounts (bots) posted only 15 times and were mostly engaged. Therefore, the results proves that this high-volume automated feed attracts very little engagement relative to its output, rather than establishing a difference between authomated accounts (bots) and humans in general.

Limitations. Posts are not entirely independent: the shuffling treats each post as an independent obser- vation, however, one automated account (euvd-bot) contributed 1,499 posts, so the shuffled distribution underestimates the actual variability. Engagement is also influenced by factors follower counts and post- ing time, which this test does not control for, so the observed association does not show that being a automated account (bot) *causes* lower engagement. Furthermore, bots were identified by handle only, so automated feeds without “bot” in their name (for example cti.fyi, which posted 190 times) included in the comparison group.

2.3 Text/Content Analysis and Visualisation

Research question: What are the most frequently discussed topics in cybersecurity-related posts on Bluesky?

Cleaning and representation. We start from the 1,293 cleaned posts (Section 2.1). Posts with empty or missing text are dropped, and URLs are removed. Each post is then split into lower-case word tokens, with punctuation removed. Standard English stop words are removed, along with the two search keywords (“cybersecurity”, “infosec”, which appear in almost every post) and URL fragments (“https”, “www”, “amp”). Pure numbers are also removed. The result is a one-word-per-row table that we use for the frequency summaries here. For clustering (Section 2.4), we convert it into a TF-IDF weighted document-term matrix.

```
library(tidyverse)
library(tidytext)
library(knitr)

posts <- readRDS("data/posts_clean.rds")

custom_stop <- tibble(
  word = c("cybersecurity", "infosec",
           "https", "http", "www", "amp")
)

words <- posts %>%
  select(id, text) %>%
  filter(!is.na(text), text != "") %>%
  mutate(text = str_remove_all(text, "https?://\\S+")) %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words, by = "word") %>%
  anti_join(custom_stop, by = "word") %>%
  filter(!str_detect(word, "[0-9]+$"))

word_frequency <- words %>%
  count(word, sort = TRUE)

word_frequency %>%
  slice_head(n = 20) %>%
  ggplot(aes(x = n, y = reorder(word, n))) +
  geom_col(fill = "#6B5B95") +
  labs(x = "Frequency", y = NULL) +
  theme_minimal()

word_author_counts <- words %>%
  distinct(id, word) %>%
  left_join(posts %>% select(id, author), by = "id") %>%
  group_by(word) %>%
  summarise(authors = n_distinct(author), .groups = "drop")
```

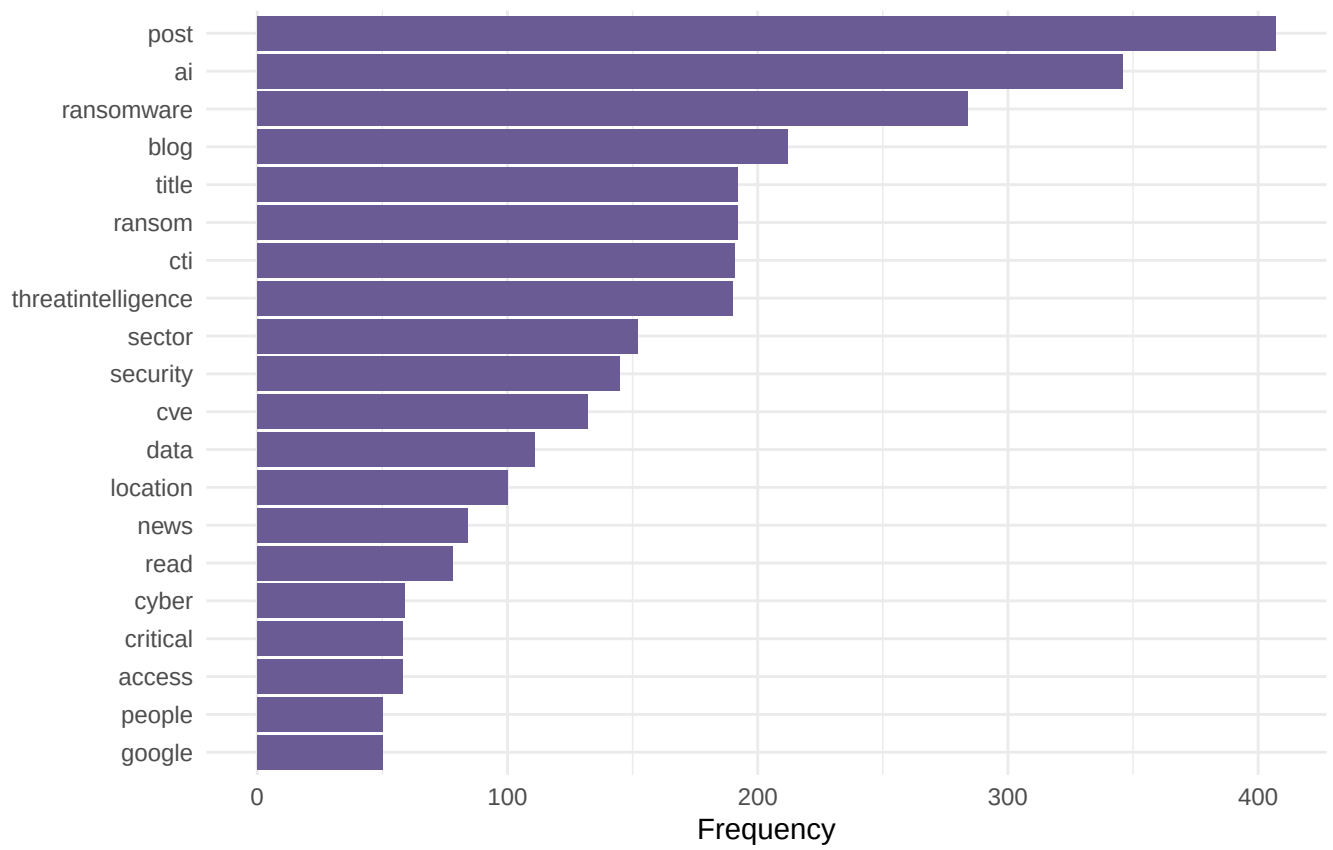


Figure 2: The 20 most frequent words in cybersecurity posts (search keywords and stop words removed)

```
word_frequency %>%
  inner_join(word_author_counts, by = "word") %>%
  slice_max(n, n = 15) %>%
  ggplot(aes(x = authors, y = reorder(word, authors))) +
  geom_col(fill = "#6B5B95") +
  labs(x = "Distinct authors", y = NULL) +
  theme_minimal()
```

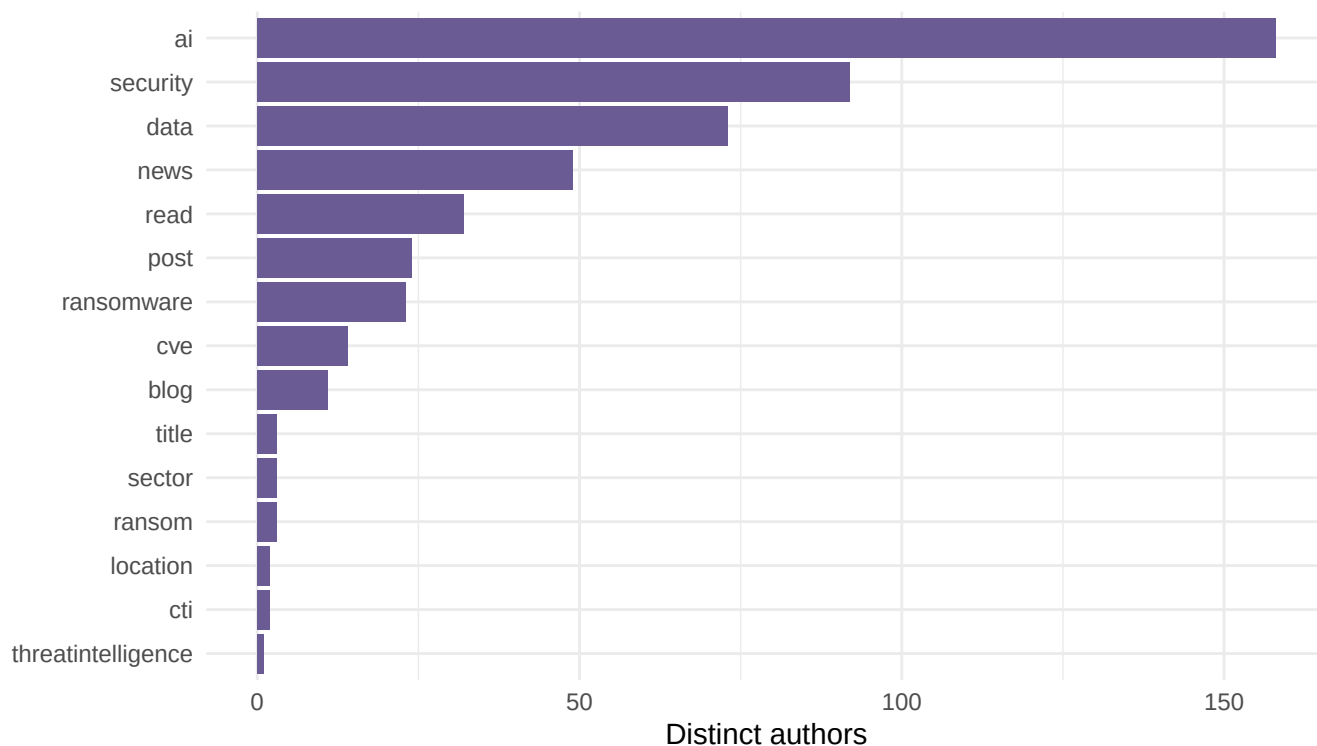


Figure 3: Number of distinct authors using each of the 15 most frequent words

Interpretation. The frequency chart shows that AI, ransomware, threat intelligence and vulnerabilities are the most prominent topics. However, counting authors changes the picture. “AI” was used by 158 different authors, so it is a topic discussed across the community. “Ransomware” came from 23 authors, and “CTI” and “threatintelligence” from only 2 and 1. A few highly active accounts can therefore make a niche topic look dominant in raw word counts. This matches the automated feeds noted in Sections 2.1 and 2.2, such as cti.fyi. Repeated meta-words like “post”, “blog” and “title” also suggest templated posts from news feeds. Because frequency alone mixes broad discussion with high-volume posting, Section 2.4 uses clustering to separate the main types of post.

2.4 Clustering

Research question: What groups of similar topics can be identified in cybersecurity-related Bluesky posts?

Decisions.

- *Representation:* each post is represented as a TF-IDF weighted document-term matrix, built from the cleaned tokens in Section 2.3. TF-IDF gives less weight to words that appear in many posts

and more weight to words that distinguish a post. To reduce noise and sparsity, we keep only the 500 most frequent terms.

- *Distance*: posts differ greatly in length, so we use cosine distance, which compares the direction of word profiles rather than their magnitude. Each post vector is scaled to unit length. After that scaling, the Euclidean distance used by k-means ranks posts the same way as cosine distance.
- *Number of clusters*: k is chosen with the elbow method. We plot the total within-cluster sum of squares for k = 2 to 10 and pick the point where adding more clusters stops giving large improvements.

```
library(tidyverse)
library(tidytext)
library(knitr)

posts <- readRDS("data/posts_clean.rds")

custom_stop <- tibble(
  word = c("cybersecurity", "infosec",
           "https", "http", "www", "amp")
)

cluster_words <- posts %>%
  select(id, text) %>%
  filter(!is.na(text), text != "") %>%
  mutate(text = str_remove_all(text, "https?://\\S+")) %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words, by = "word") %>%
  anti_join(custom_stop, by = "word") %>%
  filter(!str_detect(word, "[0-9]+$"))

tfidf_data <- cluster_words %>%
  count(id, word) %>%
  bind_tf_idf(word, id, n)

# Keep the 500 most frequent words to reduce noise
top_terms <- cluster_words %>%
  count(word, sort = TRUE) %>%
  slice_head(n = 500) %>%
  pull(word)

# TF-IDF document-term matrix (posts in rows, terms in columns)
tfidf_matrix <- tfidf_data %>%
  filter(word %in% top_terms) %>%
  select(id, word, tf_idf) %>%
  pivot_wider(names_from = word, values_from = tf_idf, values_fill = 0)

post_ids <- tfidf_matrix$id
X <- as.matrix(tfidf_matrix %>% select(-id))

# Remove posts with no retained terms
```

```

keep <- rowSums(X^2) > 0
X <- X[keep, , drop = FALSE]
post_ids <- post_ids[keep]

# Unit-length rows, so Euclidean distance behaves like cosine distance
X <- X / sqrt(rowSums(X^2))
dim(X)

```

```
## [1] 1261 500
```

```

set.seed(123)
k_values <- 2:10
wss <- sapply(k_values, function(k) {
  kmeans(X, centers = k, nstart = 10, iter.max = 100)$tot.withinss
})
plot(k_values, wss, type = "b", pch = 19,
     xlab = "Number of clusters (k)",
     ylab = "Total within-cluster SS")

```

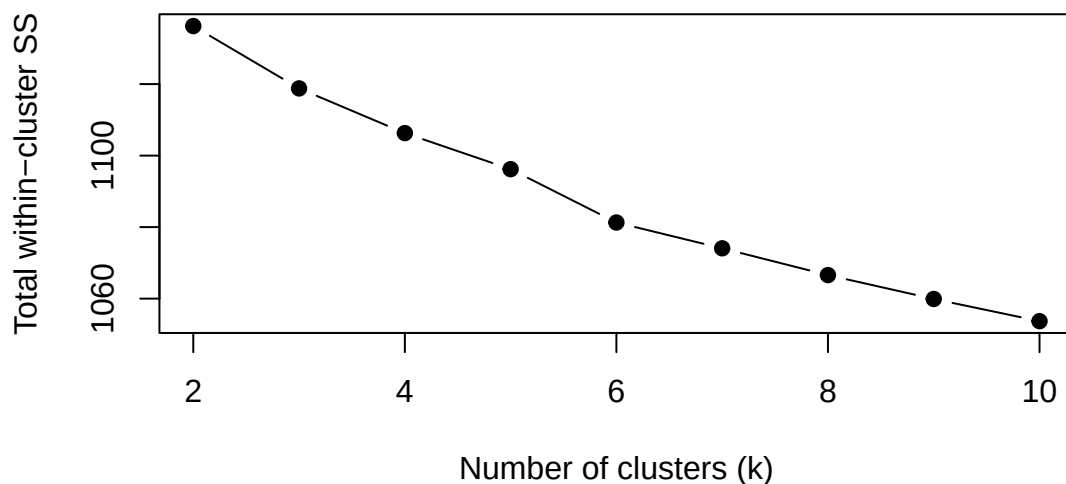


Figure 4: Elbow plot: total within-cluster sum of squares for $k = 2$ to 10

The elbow is gradual rather than sharp, but the curve starts to flatten at about $k = 6$, so we use six clusters.

```

set.seed(123)
k <- 6
km <- kmeans(X, centers = k, nstart = 25, iter.max = 100)

# Relabel clusters by size, largest first

```

```

cluster_sizes <- sort(table(km$cluster), decreasing = TRUE)

cluster_order <- as.integer(names(cluster_sizes))

new_labels <- setNames(
  seq_along(cluster_order),
  cluster_order
)

cluster_assignments <- tibble(
  id = post_ids,
  cluster = factor(
    new_labels[as.character(km$cluster)],
    levels = 1:6
  )
)

cluster_terms <- tfidf_data %>%
  inner_join(cluster_assignments, by = "id") %>%
  group_by(cluster, word) %>%
  summarise(
    mean_tfidf = sum(tf_idf) /
      n_distinct(cluster_assignments$id[cluster_assignments$cluster == first(cluster)]),
    .groups = "drop"
  ) %>%
  group_by(cluster) %>%
  slice_max(mean_tfidf, n = 10) %>%
  arrange(cluster, desc(mean_tfidf))

cluster_assignments %>%
  count(cluster, name = "posts") %>%
  mutate(share = scales::percent(posts / sum(posts), accuracy = 1)) %>%
  left_join(cluster_terms %>%
    group_by(cluster) %>%
    slice_head(n = 6) %>%
    summarise(top_terms = paste(word, collapse = ", ")),
    by = "cluster") %>%
  kable(caption = "Size and top TF-IDF terms of each cluster")

```

Table 4: Size and top TF-IDF terms of each cluster

cluster	posts	share	top_terms
1	780	62%	data, security, hacking, people, news, stay
2	191	15%	ai, openai, read, security, google, agents
3	190	15%	post, threatintelligence, cti, ransom, title, blog
4	52	4%	cve, chromium, debian, confusing, irrelevant, stackflag.com
5	29	2%	cyberangriff, teaminfosec, einbruch, angriff, datendiebstahl, sicherheitsvorfall
6	19	2%	job, jobs, educativ.net, senior, urge, hcmc

```
mds <- cmdscale(dist(X), k = 2)
plot(mds, col = as.integer(cluster_assignments$cluster), pch = 19, cex = 0.5,
     xlab = "Dimension 1", ylab = "Dimension 2")
legend("topright", legend = paste("Cluster", 1:k), col = 1:k,
     pch = 19, cex = 0.7)
```

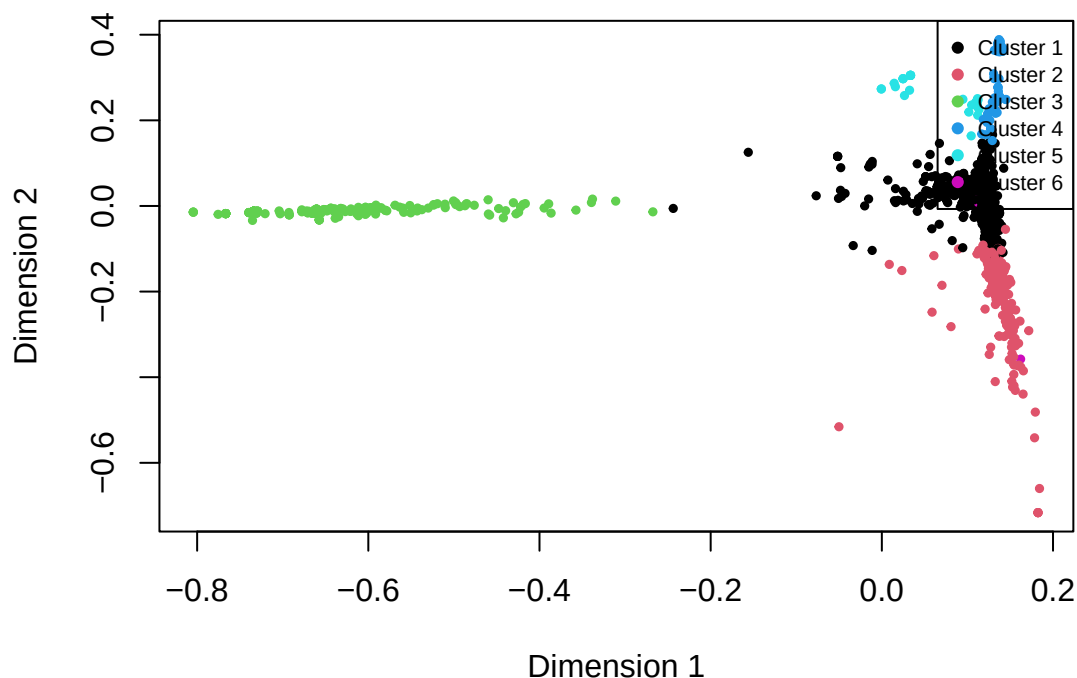


Figure 5: Posts projected into two dimensions with classical MDS on the cosine-based distances, coloured by cluster

```
cluster_terms %>%
  group_by(cluster) %>%
  slice_max(mean_tfidf, n = 8) %>%
  ungroup() %>%
  ggplot(aes(x = mean_tfidf,
             y = reorder_within(word, mean_tfidf, cluster),
             fill = cluster)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~cluster, scales = "free_y", ncol = 2) +
  scale_y_reordered() +
  labs(x = "Mean TF-IDF", y = NULL) +
  theme_minimal()
```

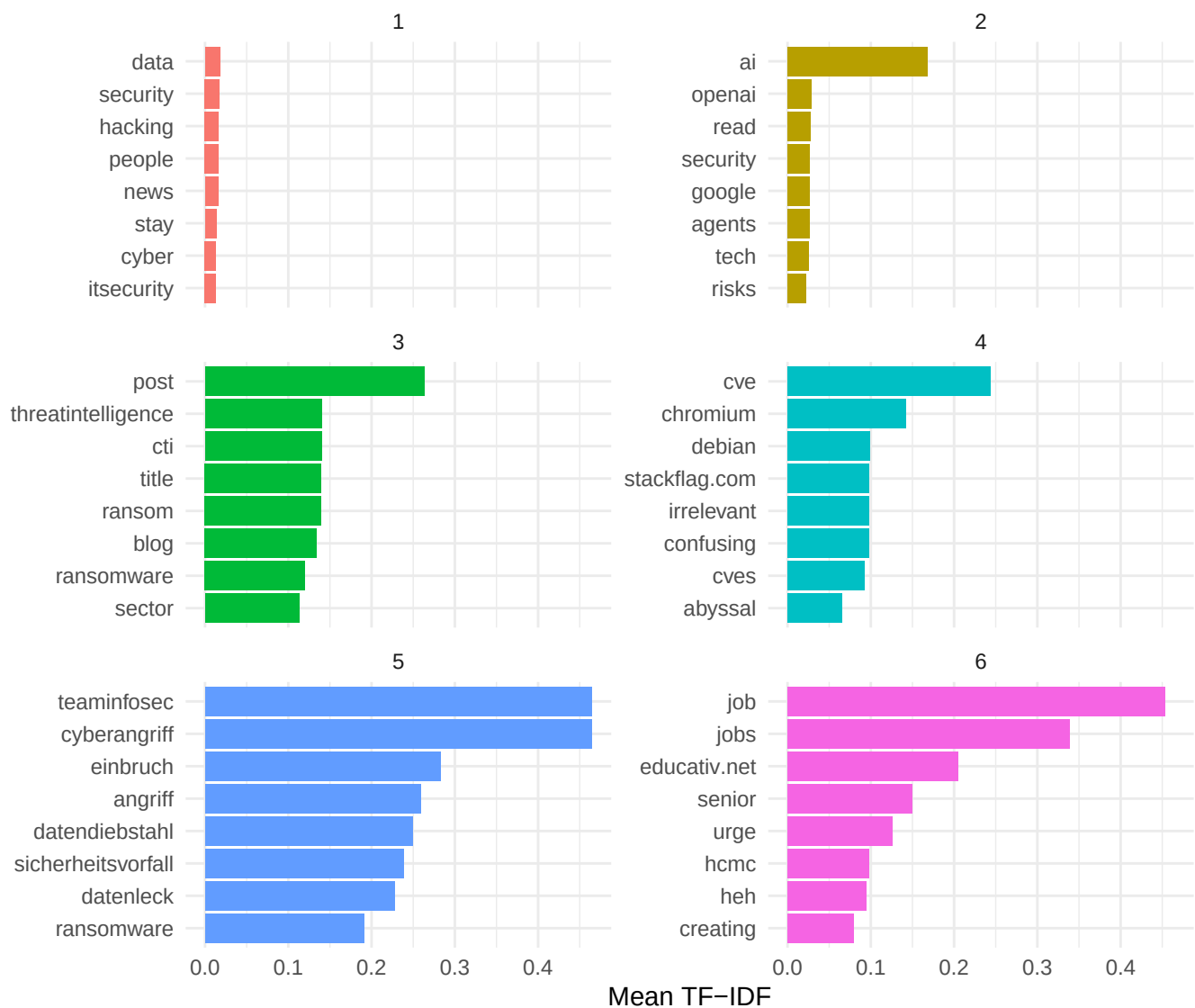


Figure 6: Top words characterising each cluster (mean TF-IDF)

Interpretation of Clusters

The K-means analysis identified six groups of cybersecurity-related posts, ranging from general discussions to specialised security alerts and employment opportunities.

Cluster 1 — General Cybersecurity Discussion (780 posts): This was the largest cluster, representing approximately 61.9% of clustered posts. Its most characteristic terms included *data*, *security*, *hacking*, *people* and *news*. The cluster reflects broad cybersecurity conversations rather than a single specialised topic.

Cluster 2 — AI and Security News (191 posts): This cluster was characterised by terms such as *AI*, *OpenAI*, *security* and *Google*. It captures discussions about artificial intelligence, emerging technologies and their implications for cybersecurity. This finding also supports the text analysis in Section 2.3, where AI was one of the most frequently discussed topics across a broad range of authors.

Cluster 3 — Ransomware and Threat-Intelligence Alerts (190 posts): Terms including *CTI*, *threatintelligence*, *ransom*, *post* and *title* dominated this cluster. These terms suggest structured threat-intelligence reporting and repeated ransomware-related alerts, rather than general conversation.

Cluster 4 — Software Vulnerabilities and CVE Alerts (52 posts): The distinctive vocabulary included *CVE*, *Chromium* and *Debian*. This cluster primarily concerns software vulnerabilities, security advisories and technical exploit information.

Cluster 5 — German-Language Cyber Incidents (29 posts): Words such as *Cyberangriff*, *Angriff*, *Einbruch* and *Datendiebstahl* indicate German-language reporting about cyberattacks, intrusions and data theft. This reveals a limitation of the English-language filtering process, as some non-English content remained in the dataset.

Cluster 6 — Cybersecurity Employment Opportunities (19 posts): This cluster contained terms such as *job*, *jobs*, *senior* and *educativ.net*, indicating cybersecurity recruitment advertisements and professional opportunities.

Overall, the clustering analysis demonstrates that cybersecurity-related posts on Bluesky are not homogeneous. Although general discussion accounts for the largest proportion of posts, the smaller clusters reveal specialised themes, including AI, vulnerability alerts, ransomware intelligence, cyber incidents and recruitment.

The results also illustrate how repeated content from a small number of accounts can influence the apparent prominence of particular topics. **Limitations.** The elbow is not sharp, so other values of k could be argued for. K-means also depends on the random starting points, and we reduce this by using 25 starts and a fixed seed. Keeping only the 500 most frequent terms may merge rarer topics into the large general cluster.

2.5 Network Analysis

Research question. *In the follow network around the most-followed active author in the cybersecurity discussion, which accounts are most central, what kind of accounts are they, and does this network belong to the cybersecurity community?*

Why a follow network rather than a mention network

A natural choice would be a weighted mention network, in which an edge $A \rightarrow B$ means that A mentioned B. However, our data cannot support a mention network. Only 40 of the 1,293 cleaned posts (3.1%) contain

a mention, and they contain 52 mentions in total. A mention network would therefore have at most 52 edges spread across a few dozen posts. It would be too small and fragmented to measure centrality in a meaningful way. Bluesky also stores mentions as account IDs (DIDs) rather than handles, so the cleaned data keeps only the mention count. We therefore follow the Module 7 seed-user method and analyse a **follow network**, which the API returns directly. The two networks answer different questions. A mention network shows who talks to whom in cybersecurity posts. A follow network shows whose content an account has chosen to receive. All centrality results below are therefore about follow relationships, not about mention-based conversation.

Building the network

The seed is sifu.tweety.fish, the author with the most followers among authors who posted about the topic at least three times. We collected the 59 accounts the seed follows (its *friends*) and then the accounts each friend follows. 0 friend retrievals failed. All relationships come from the Bluesky API, so none are inferred by us.

- **Nodes** are Bluesky accounts.
- **Edges** are follow relationships. An edge $A \rightarrow B$ means “A follows B”.
- The graph is **directed** because following is one-way: A can follow B without B following A.
- The graph is **unweighted** because a follow either exists or does not. Unlike a mention, it has no count attached to it.

```
edges <- as.data.frame(edges_raw, stringsAsFactors = FALSE)
g <- graph_from_data_frame(edges, directed = TRUE)

# Role of each account in the sample
V(g)$role <- case_when(
  V(g)$name == seed_user_handle ~ "Seed",
  V(g)$name %in% friends_handles ~ "Friend (followed by seed)",
  TRUE ~ "Friend of friend"
)

# Link to the post data: did this account post about cybersecurity in our sample?
authors <- setdiff(unique(posts$author), "handle.invalid") # unresolved handle, not a real account
V(g)$cyber_author <- V(g)$name %in% authors
```

Network properties

```
comp_weak <- components(g, mode = "weak")
comp_strong <- components(g, mode = "strong")
tot_deg <- degree(g, mode = "all")
out_deg <- degree(g, mode = "out")
in_deg <- degree(g, mode = "in")

# Largest weakly connected component
gc <- induced_subgraph(g, comp_weak$membership == which.max(comp_weak$csizes))
```

Table 5: Properties of the follow network

Property	Value
Nodes (accounts)	3,201
Edges (follows)	3,619
Density	0.00035
Weakly connected components	1
Largest weakly connected component (nodes)	3,201
Strongly connected components	3,176
Largest strongly connected component (nodes)	24
Reciprocity (share of follows returned)	0.014
Mean total degree	2.26
Diameter (longest shortest path, directed)	4

The network has 3,201 accounts and 3,619 follow edges. Its density is 0.00035, which means only a tiny fraction of the possible connections exist. This is typical of social networks: each account follows only a handful of the thousands of others in the sample. Ignoring direction, the network forms a single weakly connected component, so the largest weakly connected component contains all 3,201 accounts. This is expected from the collection method, because every account was reached from the seed. Once direction is taken into account, the network is far less connected: there are 3,176 strongly connected components, and the largest has only 24 accounts. Only 1.4% of follows are returned, so attention in this network flows mostly one way.

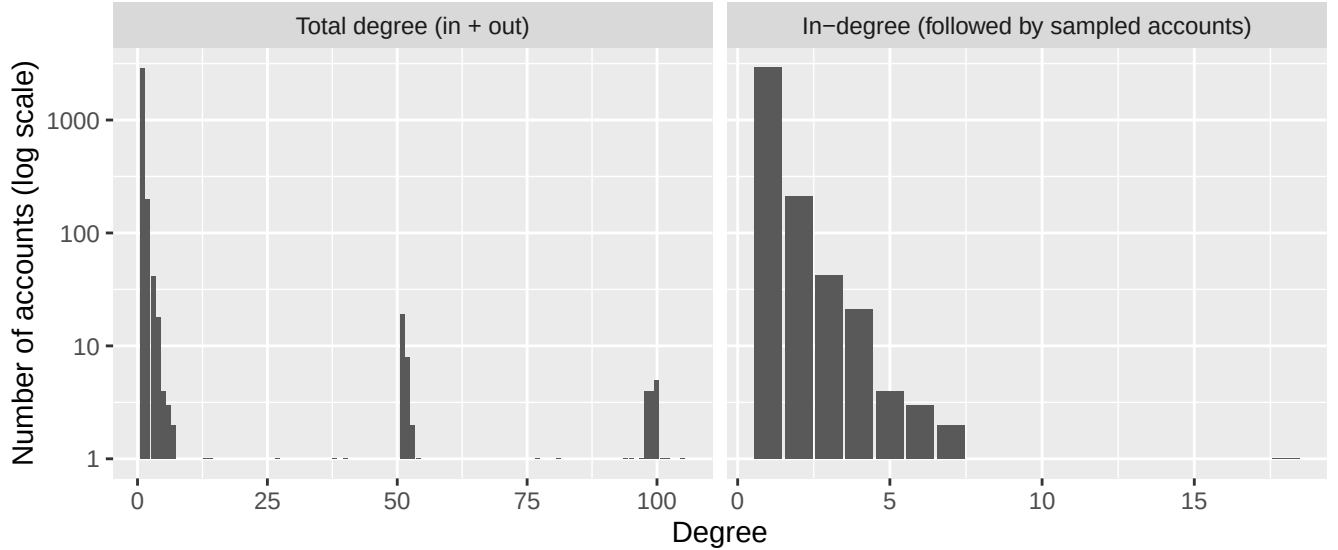


Figure 7: Degree distributions of the follow network: total degree (left) and in-degree (right). Count axis on a log scale

Both distributions are highly skewed. In the **total-degree** distribution, 2,880 accounts (90%) have a degree of 1. The highest total degrees all belong to the seed and its friends (median 52, maximum 105). Their high degree mainly reflects the follow lists we collected, because only the seed and its friends have outgoing edges. The API returned between 1 and 99 follows per friend (median 50). The **in-degree** distribution is not affected by this collection cap, so it is the better measure of popularity. 2,917 accounts (91%) are followed by only one sampled account, while a few are followed by many. This long-tailed pattern means a small number of accounts act as hubs that many of the seed’s friends choose to follow.

Table 6: Top 10 accounts by in-degree

account	role	in-degree	betweenness	posted?
sifu.tweety.fish	Seed	18	69,077	TRUE
regimecpa.bsky.social	Friend	7	6,593	FALSE
philpax.me	Friend of friend	7	0	FALSE
daniloc.xyz	Friend of friend	6	0	FALSE
segyges.bsky.social	Friend of friend	6	0	FALSE
theophite.bsky.social	Friend of friend	6	0	FALSE
elucidating.extradimensional.space	Friend of friend	5	0	FALSE
pfrazee.com	Friend of friend	5	0	FALSE
pixelatedboat.bsky.social	Friend of friend	5	0	FALSE
sky.skymarchini.net	Friend of friend	5	0	FALSE

Centrality

We use two centrality measures that answer different questions:

- **In-degree:** how many sampled accounts follow this account. A high in-degree means the account is popular or seen as worth following within the seed’s community.
- **Betweenness:** how many shortest paths between other accounts pass through this account. A high betweenness means the account is a *bridge* that connects parts of the network that would otherwise be far apart.

```
cent <- tibble(
  account      = V(g)$name,
  role         = V(g)$role,
  in_degree    = in_deg,
  betweenness  = round(betweenness(g, directed = TRUE)),
  posted       = V(g)$cyber_author
)

top_in <- cent %>% arrange(desc(in_degree), desc(betweenness), account) %>% slice_head(n = 10)
top_btw <- cent %>% arrange(desc(betweenness), account) %>% slice_head(n = 10)
```

Interpretation. The seed, sifu.tweety.fish, ranks first by in-degree (18) and first by betweenness (69,077). Its in-degree means that 18 of its 59 friends follow it back. Part of its centrality comes from the sampling design, because every account in the network was reached through the seed. Apart from the seed, the most followed accounts are regimecpa.bsky.social, philpax.me, daniloc.xyz, segyges.bsky.social, theophite.bsky.social (in-degree 7, 7, 6, 6, 6). These are accounts that several of the seed’s friends independently chose to follow, which makes them shared reference points in this community. Of the 8 accounts in the top-5 table, 8 are individuals according to their public profiles, and none are news outlets or companies. They are mainly software, AI and academic voices. Bluesky’s chief technology officer (pfrazee.com) is also among the most followed accounts, with an in-degree of 5. After the seed, every account in the betweenness top 10 is one of the seed’s friends.

Betweenness leaders. regimecpa.bsky.social is the only account apart from the seed that ranks in the top five on both measures, so it is both widely followed and a bridge. Among the remaining accounts, the highest betweenness belongs to emilymoin.com (betweenness 4,712, in-degree 2, follows 96

Table 7: Top 10 accounts by betweenness

account	role	in-degree	betweenness	posted?
sifu.tweety.fish	Seed	18	69,077	TRUE
regimecpa.bsky.social	Friend	7	6,593	FALSE
emilymoin.com	Friend	2	4,712	FALSE
jefferyharrell.bsky.social	Friend	3	4,696	FALSE
mjcrockett.bsky.social	Friend	2	4,069	FALSE
denam.bsky.social	Friend	3	2,519	FALSE
jmiers230.bsky.social	Friend	2	2,466	FALSE
emilyrcwilson.bsky.social	Friend	4	2,359	FALSE
bill-of-lefts.bsky.social	Friend	2	2,336	FALSE
joshuagraha12.bsky.social	Friend	1	2,304	FALSE

Table 8: Who are the most central accounts? (top 5 by each measure)

account	role	in-deg.	betw.	type	profile says
sifu.tweety.fish	Seed	18	69,077	Individual	Ex-cDc hacker, ML PhD, writer
regimecpa.bsky.social	Friend	7	6,593	Individual	Accountant, satire
philpax.me	Friend of friend	7	0	Individual	Software engineer
danioloc.xyz	Friend of friend	6	0	Individual	Technologist
segyges.bsky.social	Friend of friend	6	0	Individual	AI newsletter writer
emilymoin.com	Friend	2	4,712	Individual	Physician, ethicist
jefferyharrell.bsky.social	Friend	3	4,696	Individual	Tech hobbyist
mjcrockett.bsky.social	Friend	2	4,069	Individual	Academic (psychology)

sampled accounts, 60 of which no other sampled account follows); jefferyharrell.bsky.social (betweenness 4,696, in-degree 3, follows 50 sampled accounts, 36 of which no other sampled account follows); mjcrockett.bsky.social (betweenness 4,069, in-degree 2, follows 50 sampled accounts, 40 of which no other sampled account follows). These accounts have low in-degree, so few accounts in the sample follow them, and they are not popular in the in-degree sense. Their betweenness comes from their position instead. Each one is the only route from the seed to a large set of accounts that no other sampled account follows, so removing any of them would cut those accounts off from the rest of the network. By their public profiles they are: physician, ethicist; tech hobbyist; academic (psychology). They therefore connect the seed's circle to audiences outside security rather than to security communities.

Betweenness must be read with care here. Only the seed and its friends have outgoing edges, so only these 60 accounts can lie on a path between other accounts. Betweenness therefore ranks bridges *within the sample*, not across Bluesky as a whole.

Visualising the network

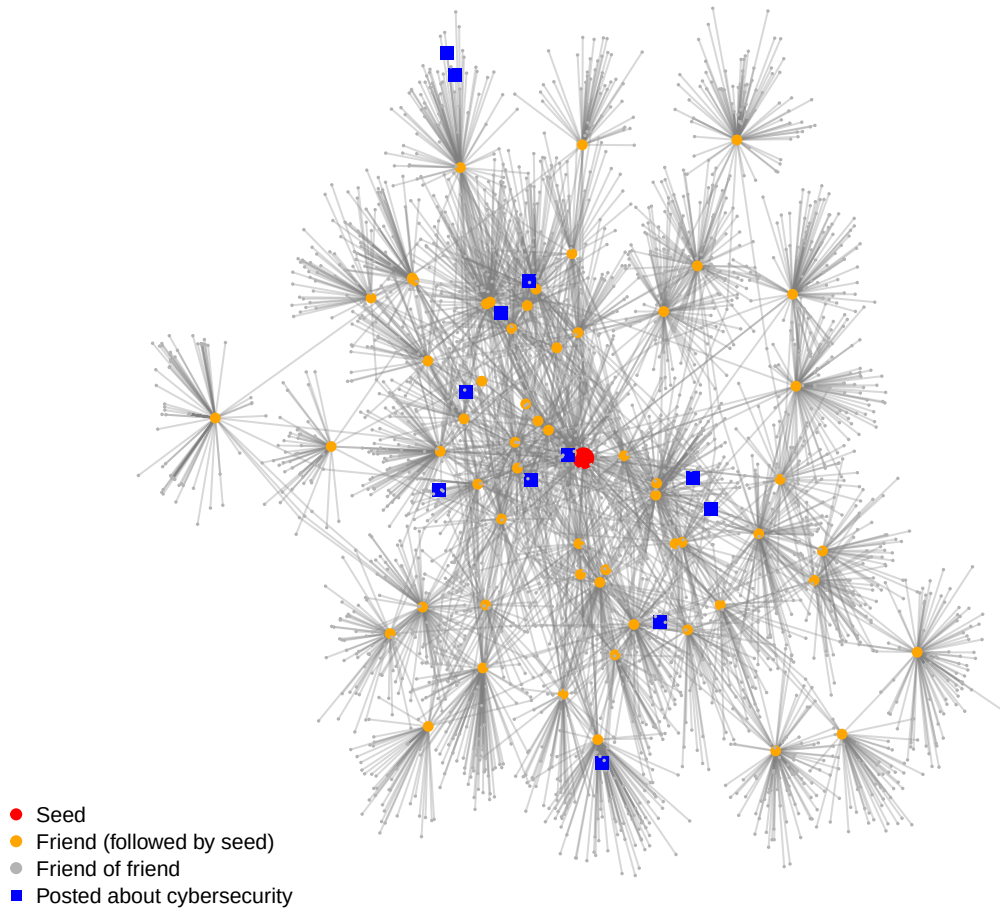


Figure 8: Largest weakly connected component of the follow network (here, the full network). Red = seed, orange = the seed's friends, grey = friends of friends, blue squares = other accounts that posted about cybersecurity in our data

The figure above shows the largest weakly connected component. Because the network has only one component, this is the whole network. It has a star-like structure: the seed sits at the centre, and each

friend has its own “fan” of followed accounts. Fans overlap only where friends follow the same accounts. With 3,201 nodes, the graph is too dense to read account names, so we also plot a subgraph.

Subgraph choice. We take the **50 accounts with the highest total degree** in the largest weakly connected component, together with the edges among them. At 50 nodes the subgraph is small enough to label. Total degree here mostly measures how many follows we *collected* for an account, so this subgraph highlights the sampled follow activity around the seed. It does not necessarily show the community’s most influential accounts. Those are better identified by in-degree (see the in-degree table above).

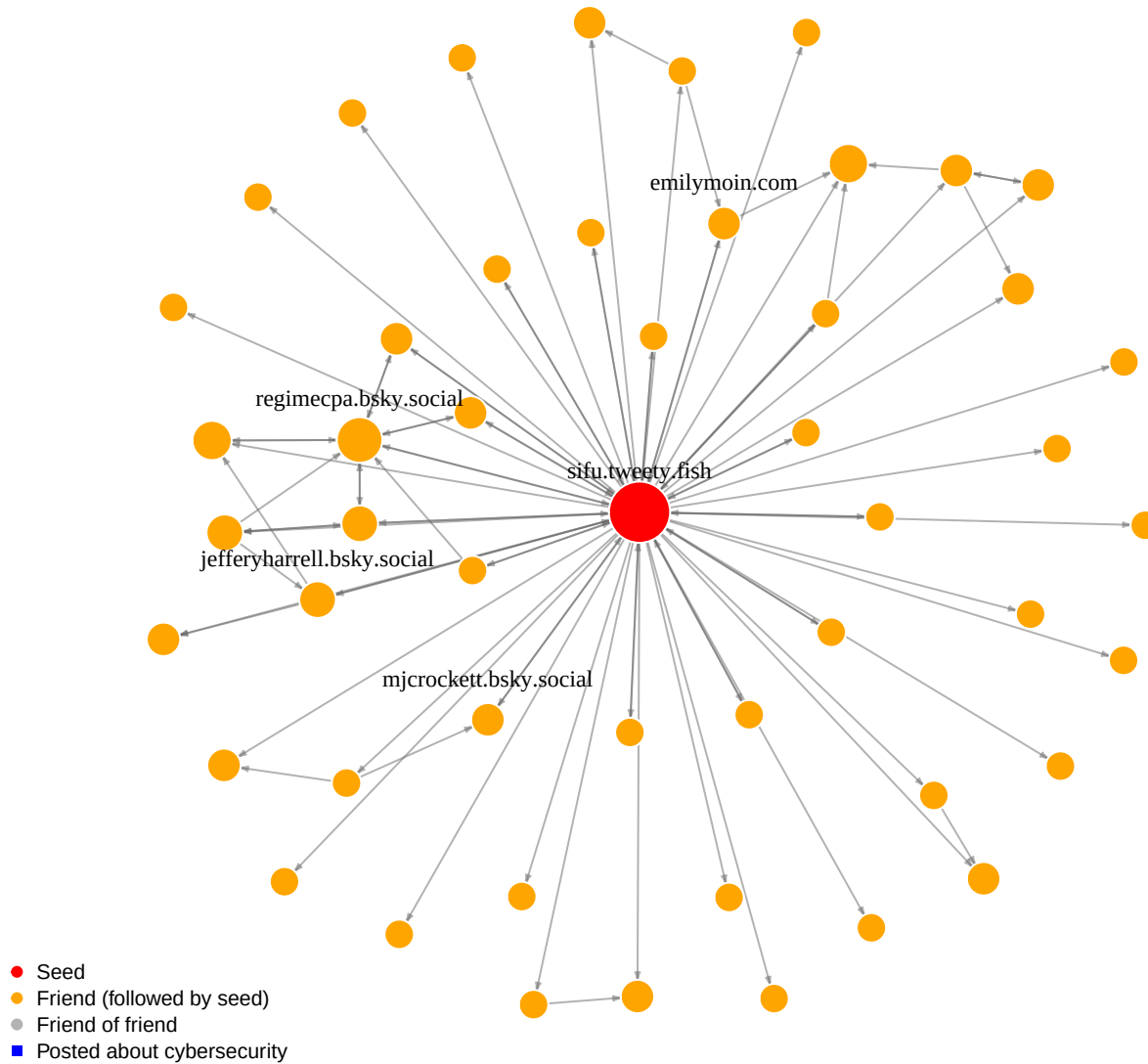


Figure 9: Subgraph of the 50 accounts with the highest total degree in the largest weakly connected component. Node size = in-degree within the full network; red = seed, orange = the seed’s friends

The subgraph contains the seed, 49 of its friends and 0 friends of friends, joined by 93 edges. Total degree is dominated by the follow lists we collected, so the 50 highest-degree accounts are all accounts whose follows were sampled, and the commonly followed hubs identified by in-degree do not appear here. Of the subgraph’s 93 edges, 49 run from the seed to its friends and 17 run from friends back to the seed. There are only 36 follow edges among the 59 friends themselves, so they are only loosely tied to one another and are held together mainly through the seed.

Table 9: Cybersecurity posts by authors inside vs outside the follow network

in_network	authors	posts	median_likes	mean_likes
FALSE	561	1273	0.0	1.8
TRUE	13	20	8.5	29.1

Link to the other analyses

```
net_authors <- intersect(V(g)$name, authors)

link_tab <- posts %>%
  mutate(in_network = author %in% net_authors) %>%
  group_by(in_network) %>%
  summarise(authors = n_distinct(author), posts = n(),
            median_likes = median(like_count), mean_likes = round(mean(like_count), 1))
ktab(link_tab, caption = "Cybersecurity posts by authors inside vs outside the follow network")
```

Only 13 of the 3,201 accounts in the network (0.4%) wrote any of the 1,293 cybersecurity posts in our sample. The seed’s profile describes a former member of the hacker group cDc and an ML researcher, but the accounts it and its friends follow are mostly general technology, AI and academic voices rather than a cybersecurity community. Posts from these 13 network members receive more engagement than other posts (median 8.5 vs 0 likes). These are only 20 posts and include the seed’s own widely liked posts, so this is a descriptive comparison, not a test. It is consistent with Section 2.2, where human accounts were far more likely to be engaged with than bots.

```
# Most common cluster (Section 2.4) for each network account that posted.
# Runs only when data/posts_clustered.rds exists; otherwise it does nothing.
if (has_clusters) {
  net_clusters <- readRDS(clustered_file) %>%
    filter(author %in% net_authors, !is.na(cluster)) %>%
    count(author, cluster) %>%
    group_by(author) %>%
    slice_max(n, n = 1, with_ties = FALSE) %>%
    ungroup() %>%
    rename(account = author, main_cluster = cluster, posts_in_cluster = n)

  ktab(net_clusters, caption = "Most common Section 2.4 cluster for network accounts that posted")
}
```

The table above links the network to the clustering in Section 2.4 by giving each posting network account its most common cluster. 2 different cluster(s) appear among these 12 accounts. Colouring the network nodes by cluster would be a natural way to link the analyses, but cluster labels exist only for accounts that posted, which is 13 of 3,201 nodes. Colouring the plots by cluster would leave more than 99% of nodes uncoloured, so we report this link as a table instead.

Of the 13 network accounts that posted about cybersecurity, 12 appear in the clustering table. The remaining account, `riziles.bsky.social`, contributed one post, but that post has no cluster assignment (NA) in the saved clustering dataset. It was therefore excluded from the cluster summary table, although it remains included in the network analysis.

Table 10: Most common Section 2.4 cluster for network accounts that posted about cybersecurity

account	main_cluster	posts_in_cluster
bdowney.bsky.social	1	4
davidho.bsky.social	2	1
dmnd.me	1	1
doctorow.pluralistic.net	2	1
hikikomorphism.bsky.social	1	1
k8em0.bsky.social	1	1
mattweiner19.bsky.social	2	1
persistentpoasting.bsky.social	1	1
sifu.tweety.fish	1	3
timkellogg.me	1	1
volts.wtf	1	1
vortexegg.com	1	1

Answer to the research question

- **Most central accounts.** The seed, sifu.tweety.fish, is the most central account on both measures, partly by design. Among the other accounts, popularity (in-degree) is held by accounts such as regimecpa.bsky.social, philpax.me, daniloc.xyz, which several of the seed’s friends follow. Bridging (betweenness) is held by friends of the seed such as emilymoin.com, jefferyharrell.bsky.social, mjcrockett.bsky.social.
- **Type of accounts.** 8 of the 8 most central accounts we reviewed are individuals, and none are news outlets or companies. Most work in software, AI or academia.
- **Is it a cybersecurity community?** No. Only 13 of 3,201 accounts (0.4%) posted about cybersecurity in our sample. The seed has a security background, but the community it follows is a general technology and AI audience. Within this sample, cybersecurity discussion on Bluesky is not concentrated in this highly followed human’s network.

Limitations

- **Follow network instead of mention network.** Mentions were too rare to build a mention network, so this analysis shows who follows whom, not who talks to whom about cybersecurity.
- **Two-step sample around one seed.** The network covers only the seed, its 59 friends and the accounts they follow, so it is an *ego network*, not the cybersecurity community on Bluesky. A different seed would give a different network.
- **Capped follow lists.** The API returned at most 99 follows per friend, so most friends’ lists are incomplete, and some edges among sampled accounts are missing.
- **Sampling shapes centrality.** Only the seed and friends have outgoing edges. This inflates their total degree and betweenness, and it means the top-50 subgraph reflects the sample design.
- **Snapshot in time.** Follows were collected once (06 October 2026), and follow relationships change over time.
- **Account types** come from reading public profile descriptions, which are self-reported and can change.

3 Overall Findings

This section answers the four research questions and then shows how the answers fit together. Engagement means at least one like, repost or reply, measured by the variables `like_count`, `repost_count` and `reply_count` at collection time.

RQ1: Text analysis (Section 2.3). *What are the most frequently discussed topics in cybersecurity-related posts on Bluesky?* Using the word counts from Section 2.3, the ten most frequent words are post (407), ai (346), ransomware (284), blog (212), ransom (192), title (192), cti (191), threatintelligence (190), sector (152), security (145). The 20 most frequent words point to ransomware and breaches, vulnerabilities, AI and threat intelligence among the main themes. Several frequent words come mainly from the ten most active accounts rather than from the wider discussion. More than half of all uses of “post”, “ransomware”, “blog”, “ransom”, “title” appear in their posts, so word frequencies are shaped strongly by a few high-volume accounts.

RQ2: Clustering (Section 2.4). *What groups of similar topics can be identified in cybersecurity-related Bluesky posts?* The clustering in Section 2.4 grouped the posts into 6 clusters: **1** (780 posts, 62%; top words: security, data, news); **2** (191 posts, 15%; top words: ai, security, openai); **3** (190 posts, 15%; top words: post, blog, cti); **4** (52 posts, 4%; top words: cve, chromium, confusing); **5** (29 posts, 2%; top words: ransomware, cyberangriff, teaminfosec); **6** (19 posts, 2%; top words: job, jobs, educativ.net). The largest cluster holds 62% of posts.

RQ3: Hypothesis test (Section 2.2). *Are posts from bot accounts less likely to receive engagement than posts from other accounts?* Yes, in this sample. A chi-squared test of independence with a simulated p-value (2,000 shuffles) rejected the null hypothesis that account type and engagement are independent ($\chi^2 = 549.55$, simulated $p = 5 \times 10^{-4}$). 56.5% of human-authored posts received at least one like, repost or reply, compared with 14.5% of bot posts, so posts from other accounts had an engagement rate about 3.9 times higher. However, 99% of the bot posts come from a single vulnerability-alert feed (euvd-bot), so the result mainly describes that feed rather than bots in general. Bots wrote 53% of the 2,845 English posts collected. This is an association, not a causal effect, because the test does not control for follower counts or posting time.

RQ4: Network analysis (Section 2.5). *In the follow network around the most-followed active author in the discussion, which accounts are most central, and is it a cybersecurity community?* The seed, sifu.tweety.fish, is the most central account, and 18 of its 59 friends follow it back. Apart from the seed, the most central accounts we reviewed are individuals working in software, AI and academia rather than news outlets or companies. The sampled follow network does not appear to be predominantly cybersecurity-focused. Only 13 of its 3,201 accounts (0.4%) posted about cybersecurity in our sample. Because mentions were too rare to build a mention network, this answer is about follow relationships.

How the analyses connect

Taken together, the four analyses describe a cybersecurity discussion on Bluesky that is high-volume but substantially automated. Section 2.1 found that bots wrote 53% of the English posts. Even after they were removed, the ten most active accounts, mostly alert and news feeds, still wrote 32% of the cleaned posts. The text analysis shows the same pattern, since 11 of the 20 most frequent words come mainly from these accounts. In the clustering, cluster 5 has the highest share of posts from these accounts (100%) and cluster 6 the lowest (0%), so the clusters partly separate feed-style content from more conversational posts. The hypothesis test shows that bot posts, almost all from one alert feed, attracted less engagement: posts from other accounts were about 3.9 times as likely to be engaged with. The network analysis adds that the most-followed active author we sampled sits in a general technology and AI community. Its network members wrote only 20 of the 1,293 posts, but those posts had a median of 8.5 likes compared

with 0 for other posts. **Main conclusion:** the results suggest that, in this sample, automated accounts contributed substantially to the volume of cybersecurity posts on Bluesky. Human-authored posts were more likely to receive engagement, and the most-followed human voice we sampled belongs to a network that extends beyond the security field.

Overall limitations

- **Short, keyword-based sample.** The data cover about five days of posts matching “cybersecurity” or “infosec”. The results reflect that week’s news, and posts on the topic that use neither keyword are missed.
- **English only.** Posts not tagged as English were removed, and some posts in other languages carry an English tag.
- **Imperfect bot detection.** Bots were identified by handle and manual review. Automated feeds without “bot” in their name remain in the cleaned data and affect the word counts and engagement results.
- **Engagement at collection time.** Newer posts had less time to gather likes, reposts and replies.
- **One-seed follow network.** The network is a two-step sample around a single account with capped follow lists. It shows who follows whom, not who talks to whom.

Reproducibility

All code and data are in the project repository: <https://github.com/Yousuf963/comp3020-group32>. The data were collected once (`scripts/Collect.R`), cleaned once (`scripts/Clean.R`) and saved. Every section of this report loads the same saved files, and this section recalculates its summary statistics from them.